



YASHU SHUKLA

Data Science · AI Research Engineer

Department of Data Science and Cognitive Science

Hanyang University, South Korea

CONTACT

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ABOUT ME

I am an AI Research Engineer passionate about Physical AI and dexterous robotics. As a Data Science graduate from Hanyang University, my work sits at the frontier of Robot Foundation Models and Vision-Language-Action systems; spanning 3D scene-aware perception, tactile-torque sensory integration, and touch-aware dexterous manipulation policies. Beyond robotics, I bring strong hands-on experience in machine learning, NLP, and statistical modeling, having built and deployed ML systems across AI robotics labs and IT startups; from patent-filed anomaly detection to sentiment analysis at scale. I am driven to push the boundaries of what intelligent systems can do in the physical world.

EDUCATION

Bachelors of Engineering in Data Science | Hanyang University, South Korea 2022 - 2026

Major: Data Science Engineering and Physiological Brain Science

GPA: 4.0 out of 4.5

Middle and High School | Vibgyor High School, India 2015 - 2022

Major: Science (Mathematics, Physics, Chemistry, English, Physical Education)

- 2022- Ranked in the Top 5 in Grade 12 (92.6%)
- 2020- Ranked in the Top 10 in Grade 10 (93.5%)

Primary School | International School of Koje, South Korea 2008 - 2015

- Studied in an international environment
 - Experience of solving practical case studies
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TECHNICAL SKILLS

Languages: Python, C, C++, R, JavaScript, Typescript

Frameworks / Tools: TensorFlow, PyTorch, scikit learn, Keras, NumPy, pandas, Matplotlib, Seaborn, Django

Robotics & Simulation: Isaac Sim/Lab, Robot Manipulation, 3D Object Scanning, ROS

Development & Analytics Tools: Angular 10, ELK Stack, PsychoPy, API development

Databases: SQL, PostgreSQL

Cloud & Deployment: AWS, GCP

Productivity Tools: MS Office

EXPERIENCE

1) AI Research Engineer Intern - RLWRLD, Seoul

Dec 2025-Present

- Built and curated robotic manipulation datasets in Isaac Sim, optimized dataset composition to achieve equivalent model accuracy with 50% reduced data volume.
- Fine-tuned and systematically benchmarked multiple VLA models, establishing performance baselines for robot foundation model training.
- Developed end-to-end automated training and evaluation pipeline reducing model setup time from 20–30 minutes to under 30 seconds.
- Integrating 3D geometry perception into VLA models via foundation model-based scene understanding, enabling multimodal sensor fusion (tactile, joint torque) for touch-aware dexterous manipulation policies.

2) Data Scientist - w3company, Seoul

March 2025-Dec 2025

(Part-time)

- Part of the R&D team developing machine learning-driven multi-module system for fake account detection and anomaly detection; (patent filed)
- Built and evaluated deep learning image classifiers using Residual Networks (ResNet) for synthetic image detection, targeting AI-generated profile photos and fraudulent user identities
- Developed and evaluated BERT-based NLP models for sentiment analysis and text classification to support improvements in a social networking app (Moji) and provide user engagement insights

3) Undergraduate Thesis Research

March 2025-Dec 2025

(Advisor: Prof. Noh Youngtae, Department of Data Science)

- Research aimed to enable non-invasive continuous health monitoring by addressing noise interference in mmWave FMCW radar (AWR1642BOOST) vital sign signals.
- Developed a CNN+BiLSTM+Attention model for contactless heart rate and breathing rate estimation from mmWave radar signals; 3× improvement over classical signal processing baselines.

4) Teaching Assistant - Hanyang University, Seoul

Sep. 2024-Dec. 2024

- Managed and tracked a class of over 30 students for the course “Understanding Business Environment
- Helped facilitate group discussions and ensured smooth execution of class activities

5) Software Engineer Intern - Bespoke System Pvt. Ltd., India

July 2024-Aug. 2024

- Worked on web development using ASP.NET framework, MS SQL, Angular 10, Web API, and SQL Server
 - Gained experience in both frontend and backend development, focusing on API integration and database management
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PUBLICATION & CONFERENCES

- 1) Research Contributor to the “RLDX-1 Technical Report” ([arXiv](#))

Seoul, May 2026

Dongyoung Kim, Myungkyu Koo, ... **Yashu Shukla**, ... Yeonjae Lee, Jinwoo Shin

- Contributed to research on Robot Foundation Models and Vision-Language-Action (VLA) systems for dexterous humanoid robot manipulation
 - Developed an end-to-end automated pipeline for VLA model training and inference
- 2) The 12th Swiss-Korean Life Science Symposium, Research Attendee

Seoul, May 2025
- 3) Presented research poster, “Cerebellum Activation in Depression” at the Korean Society for Brain and Neural Sciences ([KSBNS](#)) Conference

Seoul, Aug. 2025
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PROJECTS

- 1) Contactless Vital-Sign Monitoring Using mmWave Radar

March 2025 - Dec 2025

- Benchmarked three denoising frameworks (BPF+FFT, DWT, CNN+BiLSTM+Attention) for mmWave radar heart rate and breathing rate estimation.
 - Deep learning model achieved MAE of 3.12 (heart rate) and 2.99 (breathing rate), significantly outperforming classical signal processing baselines; enabled driver stress detection and clinical health monitoring applications
- 2) StressMate – Mental Wellness Web App

May 2025 - June 2025

- Built a Django-based app integrating LLMs (Perplexity AI, Gemini) along with RAG
 - Deployed on AWS using EC2 and S3 for scalable hosting and cloud based storage
- 3) Cerebellar fMRI Analysis in Major Depressive Disorder

May 2025 - June 2025

- Conducted advanced fMRI analysis revealing significantly reduced cerebellar response to positive stimuli in clinical depression patients compared to neurotypical participants, highlighting emotion-processing deficits
- 5) PlateCheck Application

Oct. 2024 - Dec. 2024

- Developed LLM based Korean Food Ingredient Recognition that accommodates both food image and text inputs using Django
 - Integrated a fine-tuned ResNet model for image-based dish identification and utilized GPT-4o Mini for extracting relevant ingredient details.
 - Conducted and evaluated the service based on HCI factors via user study
- 6) Food Demand Forecasting using Temporal Convolutional Networks

Oct. 2024 - Dec. 2024

- Built a time series forecasting model using Temporal Convolutional Network (TCN) to forecast food demand based on historical sales data
- 7) News Audio Summarizer using BART

April 2024 - June 2024

- Designed an NLP-based system for converting audio news content to text and generating concise summaries by fine tuning the BART model on transcribed news articles paired with human written summaries
- 9) Sign Language Recognition Model with MLP and CNN

Nov. 2023

- Developed a machine learning model to recognize sign language gestures, leveraging both Multi-Layer Perceptrons (MLP) and Convolutional Neural Networks (CNN)
- 10) Clothing Size Prediction Model for Fashion E-Commerce

April 2022 - June 2022

(Group Project)

- Developed Clothing Size Prediction Model for Fashion E-Commerce with Machine Learning Approaches
 - Collected data from the MUSINSA online clothing store using Python Selenium
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LANGUAGES

- **Native:** English and Hindi
 - **Intermediate:** Gujarati
 - **Beginner:** Korean
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SCHOLARSHIPS AND ACHIEVEMENTS

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| • Excellence Award, College of Engineering, Hanyang University | August 2026 |
| • Hanyang Brain (Academic Excellence) Scholarship | March 2026 - June 2026 |
| • Daewoong Foundation Global Scholarship | Nov. 2025 - March 2026 |
| • Hanyang Brain (Academic Excellence) Scholarship | Sep. 2025 - Dec. 2024 |
| • Hanyang Brain (Academic Excellence) Scholarship | March 2025 - June 2025 |
| • Special Scholarship for Academic Encouragement Hanyang (Foreigner) | Sep. 2024 - Dec. 2024 |
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EXTRACURRICULAR

- Maintains a personal blog on achieving balanced work and lifestyle
- Completed a 5 year Senior Diploma in Kathak (Indian Classical Dance)
- Kathak Performer – Indian Embassy Cultural Events, South Korea
- Engaged in team sports including basketball and badminton